

PROPERTY MARKET ANALYSIS

New York City Real Estate Market Analysis Using BigQuery, SQL, Python, and Power BI

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Project Domain: Real Estate

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ABSTRACT

This project investigates the New York City (NYC) residential property market using the Google BigQuery public real estate dataset. The study focuses on understanding how property prices vary across boroughs, neighbourhoods, building categories, and time periods, while identifying the borough with the strongest and most consistent long-term property price growth.

The analysis was conducted using SQL in Google BigQuery, Python for exploratory data analysis (EDA), and Power BI for dashboard visualisation. Data cleaning procedures were applied to remove invalid and non-market transactions, while the analysis focused specifically on residential property categories.

Key metrics examined include:

- average sale price,
- median sale price,
- price per square foot,
- transaction volume,
- and year-over-year (YoY) growth.

The findings provide insight into:

- borough-level affordability,
- neighbourhood price segmentation,
- market appreciation trends,
- and property value density across New York City.

The project concludes by answering the crown metric question:

Which NYC borough has experienced the strongest and most consistent property price growth over time?

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Real estate is one of the most important sectors in urban economics because property markets reflect broader economic activity, population growth, investment patterns, and housing demand.

Property values are influenced by several factors, including:

- location,
- neighbourhood desirability,
- property size,
- infrastructure development,
- building age,
- and market demand.

In a large metropolitan city such as New York City, the real estate market is highly dynamic and segmented. Property prices differ significantly across boroughs such as:

- Manhattan,
- Brooklyn,
- Queens,
- Bronx,
- and Staten Island.

Even within the same borough, neighbourhood-level variations can be substantial.

The availability of public real estate transaction data creates an opportunity to apply modern data analytics techniques to uncover hidden patterns and market behaviour. Using tools such as Google BigQuery, SQL, Python, and Power BI makes it possible to transform raw property sales data into meaningful business insights.

This project focuses on analysing NYC residential property transactions to understand:

- how property prices behave across boroughs,
- how values change over time,
- and which borough demonstrates the strongest market appreciation.

1.2 Problem Statement

The NYC property market contains thousands of transactions involving different:

- property types,
- price ranges,

- building sizes,
- neighbourhoods,
- and historical periods.

Raw transaction data alone does not provide meaningful insights unless it is properly cleaned, analysed, and visualised.

Without structured analysis, stakeholders may struggle to answer important market questions such as:

- Which borough is the most expensive?
- Which neighbourhoods command premium prices?
- How do building types influence property value?
- Which borough has experienced the strongest long-term growth?

This project addresses these challenges using data analytics techniques to generate structured, evidence-based insights from historical property sales data.

1.3 Aim of the Project

The main aim of this project is to analyse the NYC residential property market and determine which borough has experienced the strongest and most consistent price growth over time.

1.4 Objectives of the Study

The objectives of this study are to:

1. Extract and clean the NYC property sales dataset from Google BigQuery.
2. Analyse borough-level and neighbourhood-level pricing patterns.
3. Examine the distribution of residential property prices.
4. Compare price per square foot across boroughs.
5. Study historical property price trends.
6. Investigate the relationship between building age and sale price.
7. Compare average prices across residential building categories.
8. Develop an interactive Power BI dashboard.
9. Identify the borough with the strongest long-term price appreciation.

1.5 Research Questions

This project seeks to answer the following questions:

1. How many valid property sales remain after data cleaning?
2. What period does the dataset cover?
3. Which borough is the most expensive?
4. Which borough is the least expensive?
5. What does the distribution of property sale prices look like?
6. Which borough has the highest average price per square foot?
7. How have property prices changed over time?
8. Which neighbourhoods have the highest and lowest average sale prices?
9. Is there a relationship between building age and sale price?
10. Which borough has experienced the strongest property price growth?

1.6 Significance of the Study

This project provides valuable insights for several stakeholders.

Real Estate Investors

To identify high-growth markets and investment opportunities.

Property Buyers

To compare affordability and property value across boroughs.

Urban Planners

To understand neighbourhood growth patterns and housing demand.

Policy Makers

To monitor affordability challenges and urban development trends.

Data Analysts

To demonstrate practical applications of SQL, Python, BigQuery, and Power BI using real-world public datasets.

1.7 Scope of the Study

This study focuses exclusively on residential property transactions within the NYC public property sales dataset.

Included in Scope

- Residential property sales
- Borough comparisons
- Neighbourhood comparisons
- Historical pricing trends
- Price per square foot analysis
- Building age analysis
- Monthly transaction patterns

Excluded from Scope

- Commercial properties
- Industrial properties
- Mortgage rate analysis
- Inflation-adjusted pricing
- Macroeconomic forecasting
- Rental market analysis

CHAPTER TWO: LITERATURE AND CONCEPTUAL REVIEW

2.1 Property Market Analytics

Property market analytics involves the use of data to analyse:

- real estate pricing,
- transaction behaviour,
- market demand,
- and investment performance.

Important variables commonly analysed include:

- location,
- property size,
- building age,
- transaction volume,
- and time-based trends.

One major challenge in real estate analytics is that property prices are often highly skewed. A small number of luxury property transactions can distort average prices.

Because of this, analysts frequently use:

- median sale price,
- price per square foot,
- and distribution analysis

to obtain more reliable comparisons.

2.2 Key Real Estate Metrics Used in This Study

Average Sale Price

The mean sale price of all valid transactions.

Median Sale Price

The middle value when property prices are ranked from lowest to highest. This metric reduces the effect of extreme outliers.

Price per Square Foot

Price per Sqft=Gross Square Feet /Sale Price

This metric standardises property value relative to size.

Year-over-Year Growth

YoY Growth %=((Current Year Avg Price–Previous Year Avg Price)/ Previous Year Avg Price) × 100

This measures how average property prices change from one year to another.

Transaction Volume

The total number of property transactions within a specific period.

CHAPTER THREE: METHODOLOGY

3.1 Research Design

This project adopts a quantitative, exploratory, and descriptive analytical approach.

The study is data-driven and focuses on identifying:

- market trends,
- pricing behaviour,

- and relationships within historical NYC residential property transactions.

3.2 Data Source

The dataset used for this project was obtained from the Kaggle public dataset repository.

Primary Dataset

<https://www.kaggle.com/datasets/new-york-city/nyc-property-sales?select=nyc-rolling-sales.csv>

This dataset contains historical property transaction records across New York City.

3.3 Tools Used

Tool	Purpose
Google BigQuery	Querying large public datasets
SQL	Data extraction, filtering, and aggregation
Python	Data cleaning, EDA, and visualisation
Power BI	Dashboard development and reporting
Google Colab Notebook	Python execution environment

3.4 Data Cleaning Procedure

Several cleaning procedures were applied to ensure the dataset accurately reflected real market transactions.

Step 1: Remove Non-Market Transactions

Transactions with:

- sale_price = 0
- sale_price < 10000

were removed because they often represent:

- family transfers,
- administrative transfers,
- or non-market transactions.

Step 2: Restrict to Residential Properties

Only the following residential property categories were included:

- One Family Dwellings
- Two Family Dwellings
- Three Family Dwellings

Step 3: Handle Invalid Square Footage

For price-per-square-foot analysis:

`gross_square_feet > 0`

was required.

Step 4: Standardise Date Format

`sale_date` was converted into a usable datetime format for trend analysis.

Step 5: Remove Critical Null Values

Rows with missing:

- borough,
- neighbourhood,
- or essential transaction details

were excluded where necessary.

3.5 Analytical Workflow

The project followed the workflow below:

1. Define business and research questions
2. Access raw data from BigQuery
3. Clean and filter data using SQL
4. Export data into Python
5. Perform exploratory data analysis
6. Engineer additional features
7. Build Power BI dashboard
8. Interpret findings and answer the crown metric question

CHAPTER FOUR: DATA ANALYSIS AND SQL IMPLEMENTATION

This chapter presents the SQL queries used throughout the analysis process.

4.1 Base Cleaned Dataset Query

```
CREATE OR REPLACE TABLE `your_project.cleaned_nyc_property_sales` AS
```

```
SELECT
  county_name,
  neighborhood,
  building_class_category,
  sale_price,
  sale_date,
  gross_square_feet,
  land_square_feet,
  year_built,
  total_units

SELECT *
FROM `ecommerce-analytics01.nyc.property_sales`;

WHERE sale_price >= 10000
AND county_name IS NOT NULL
AND neighborhood IS NOT NULL
AND (
  building_class_category LIKE '01%'
  OR building_class_category LIKE '02%'
  OR building_class_category LIKE '03%'
);
```

Explanation

This query creates a cleaned residential property sales table by:

- removing invalid sales,
- restricting the dataset to residential categories,
- and ensuring important location fields are available.

CHAPTER FIVE: EXPLORATORY DATA ANALYSIS (EDA)

5.1 Introduction to EDA

Exploratory Data Analysis (EDA) was performed to understand:

- the structure of the dataset,
- pricing behaviour,

- historical trends,
- and borough-level differences.

The analysis focused on:

- price distributions,
- borough comparisons,
- neighbourhood segmentation,
- building characteristics,
- and time-series trends.

CHAPTER SIX: CROWN METRIC ANALYSIS

6.1 Crown Metric Question

Which NYC borough has experienced the strongest property price growth over time?

This is the central competitive question of the project.

6.2 Crown Metric Definition

The crown metric is based on year-over-year average property price growth by borough.

$\text{YoY Growth \%} = (\text{Current Year Avg Price} / \text{Previous Year Avg Price}) \times 100$

6.3 Evaluation Criteria

The strongest-performing borough should demonstrate:

- consistent positive growth,
- long-term appreciation,
- lower volatility,
- and sustained upward market movement.

CHAPTER SEVEN: DASHBOARD DESIGN

7.1 Dashboard Objective

The Power BI dashboard was designed to provide an interactive overview of NYC residential property market performance.

Users can:

- compare boroughs,
- analyse neighbourhood trends,
- evaluate property categories,
- and explore historical pricing movement.

7.2 Dashboard Components

KPI Cards

- Total Sales
- Average Sale Price
- Average Price per Sqft
- Most Expensive Borough

Charts

- Average Price by Borough
- Annual Price Trend
- Top 15 Neighbourhoods by Price
- Price per Sqft by Borough
- Building Type vs Average Price
- Monthly Transaction Volume

Slicers

- Borough
- Neighbourhood
- Building Type
- Sale Year
- Building Era

CHAPTER EIGHT: FINDINGS AND DISCUSSION

8.1 Borough Pricing

The analysis revealed major differences in property prices across NYC boroughs, showing that location remains one of the strongest determinants of residential property value.

8.2 Distribution Analysis

Property prices were heavily right-skewed, meaning a small number of luxury transactions significantly increased average prices.

8.3 Value Density

Price per square foot analysis provided a more standardised comparison of property value across boroughs.

8.4 Historical Growth

The NYC property market generally showed long-term appreciation, although growth patterns differed across boroughs.

8.5 Neighbourhood Segmentation

Neighbourhood-level analysis showed that borough-wide averages often conceal substantial internal variation.

CHAPTER NINE: RECOMMENDATIONS

9.1 For Investors

Focus on boroughs with:

- strong YoY growth,
- rising price per square foot,
- and healthy transaction activity.

9.2 For Buyers

Use:

- median price,
- and price per square foot

instead of average price alone when comparing affordability.

9.3 For Developers

Target neighbourhoods with:

- moderate current prices,
- but strong long-term appreciation potential.

CHAPTER TEN: LIMITATIONS AND CONCLUSION

10.1 Limitations

The study has several limitations:

- no inflation adjustment,
- no mortgage rate integration,
- possible residual outliers,
- missing square footage in some records,
- and restriction to selected residential categories.

10.2 Conclusion

This project successfully demonstrated how modern data analytics tools can be applied to analyse the New York City residential property market.

Using:

- Google Big Query,
- SQL,
- Python,
- and Power BI,

The project transformed raw public transaction data into meaningful insights about:

- borough-level pricing,
- neighbourhood performance,
- property value density,
- historical growth,
- and building characteristics.

The project also established a framework for identifying the borough with the strongest and most consistent property price growth over time.

Overall, this study provides a strong example of practical, real-world real estate analytics using public datasets and modern business intelligence tools.

